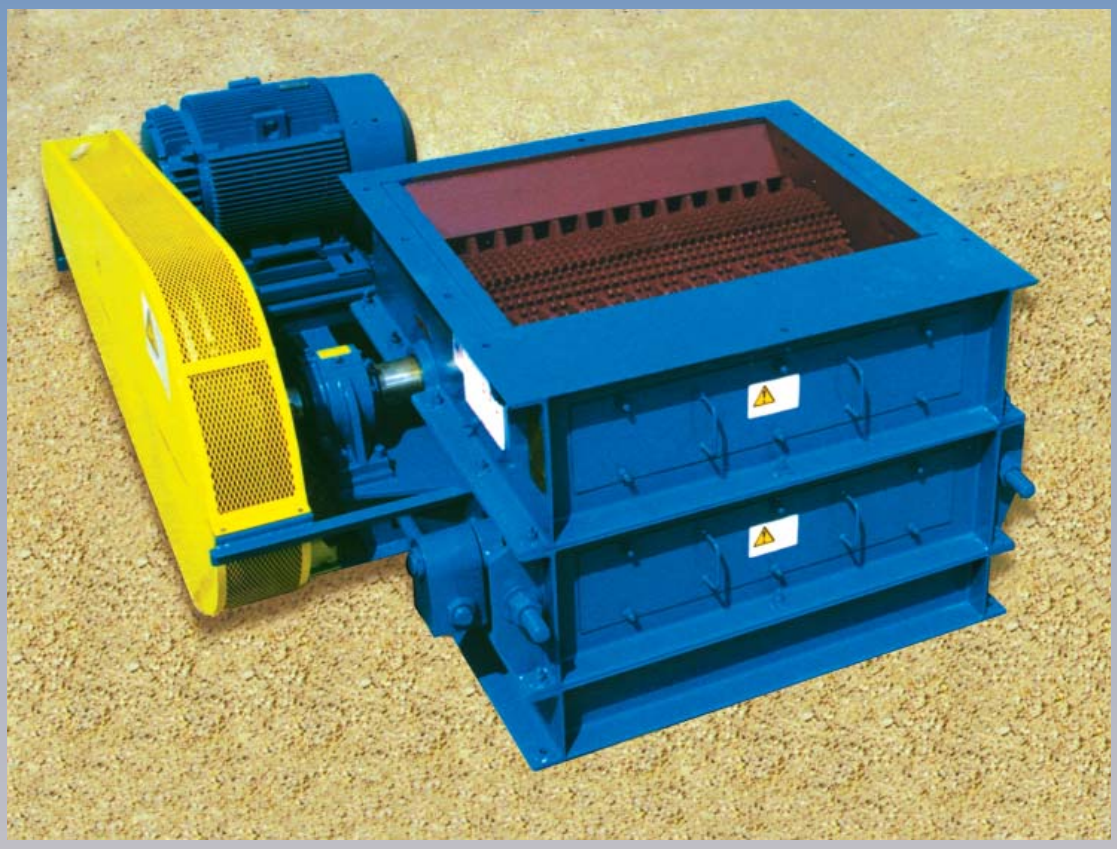
WILLIAMS

Roll Crushers

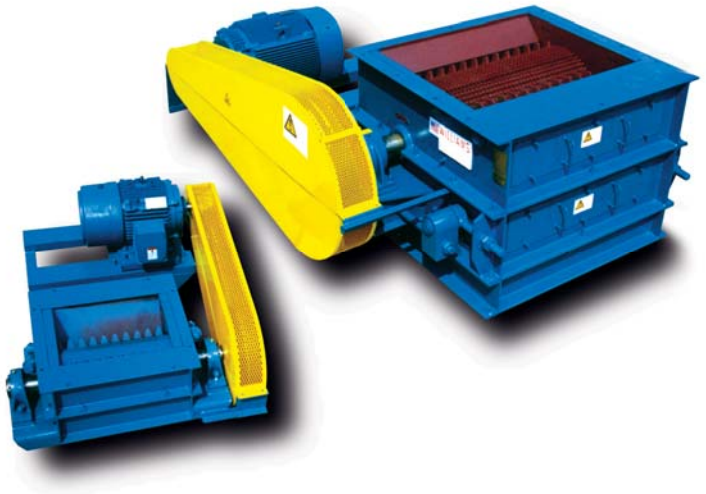
Heavy Duty Single and Double Roll Crushers



For the efficient size reduction of minerals and other friable materials.

BULLETIN 915R

Williams Roll Crushers are built rugged for high tonnage with minimum maintenance.



Roll crushers are simple in design and construction, long lasting, economical, and versatile (see list of reducible materials on back page).

Roll crushing surfaces operate at a fixed distance apart, as opposed to the continually changing distances in a jaw or cone crusher. Product size is much more consistent. Both oversized pieces and fines are minimized. Wet, sticky materials are more easily handled.

The rolls act as flywheels, contributing to smooth operation and efficient use of power.

Roll crushers are low in profile and relatively easy to install. They can be fed with a minimum of headroom, or even choke fed. Adjustments are simple and internal parts are readily accessible.

Single or Double Roll Crushers?

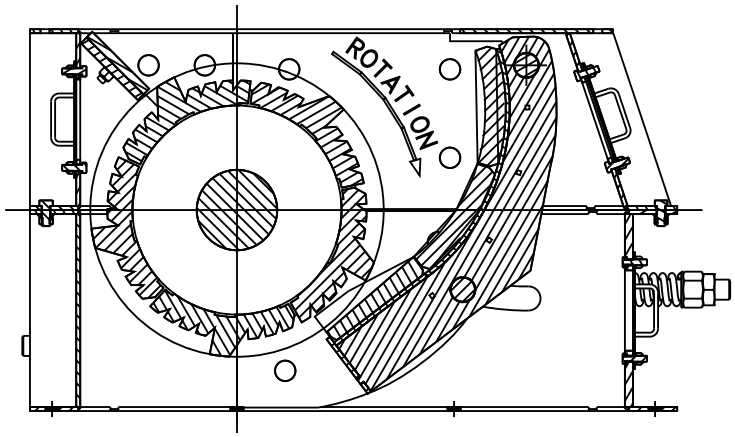
The choice depends upon the feed material, feed size, product size desired, and "consist" of both feed and product.

Single Roll Crushers reduce via a combination of impact, shear, and compression. Double Roll Crushers reduce primarily through compression, although some shear is obtained with toothed rolls. Single Rolls are always toothed in patterns suited to the feed material. Double Rolls come in combinations of smooth, corrugated, and toothed designs.

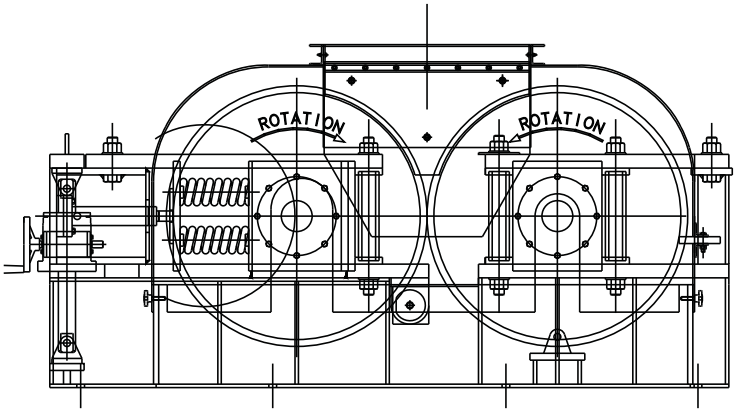
Both Single and Double Roll Crushers operated most efficiently with dry, friable materials. Single Roll Crushers, however, have been widely and successfully used for the reduction of moist clays. They also have been long used as Primary and Secondary Coal Crushers, both at mine sites and power plants, where a minimum of fines is desired.

Single Roll Crushers generally handle larger feed sizes at higher reduction rations in higher capacities. Double Roll Crushers produce a finer product at lower reduction ratios and capacities.

TYPICAL CROSS SECTION OF A SINGLE ROLL CRUSHER

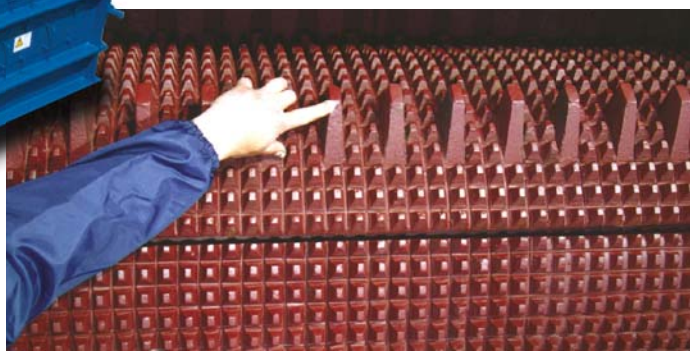


TYPICAL CROSS SECTION OF A DOUBLE ROLL CRUSHER



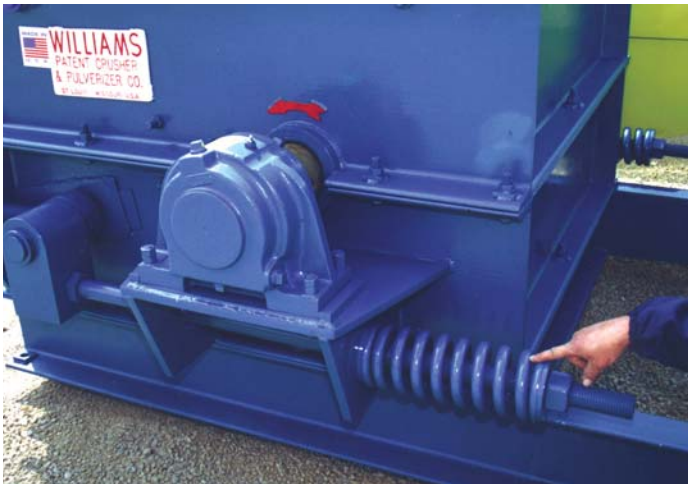


SINGLE ROLL CRUSHERS



- **Welded Steel Frame** - Heavy, rolled plate and welded structural shapes for rigidity and resistance to shock.
- **Reversible, Replaceable Liners** - Protect internal surfaces and frame. Thick, abrasion-resistant alloy. Reversible for added life. Easily replaced.
- **Custom-Engineered, Replaceable Roll Faces** - Tooth patterns and corrugations to suit feed material. Abrasion-resistant alloy. Easily replaced.
- **Protective Spring Releases** - Heavy duty compression springs permit movement of floating roll to pass tramp metal and other uncrushables, avoiding overload and damage. Smaller Single Roll Crushers are equipped with shear pin release.
- **Simple Roll Adjustment** - Quickly set and maintained via manual bolt/shim arrangement. Allows shock loads to be taken in compression rather than tension.
- **Heavy Duty Drive Shafts and Bearings** - Oversized, heat-treated, alloy steel shafts plus shaft-aligning, roller-type bearings assure long life and maximum use of power. Jackshafts for control of roller speed are standard on Double Roll Crushers, optional on larger Single Roll type.
- **Choice of Drives** - Multiple V-belt drives recommended for maximum shock absorption. Shaft-mounted speed reducers optional.
- **Easy Accessibility** - Housings for Single Roll Crushers split and open at centerline for access to all working parts. Double Roll Crusher models dismantle readily.
- **Low Peripheral Speed** - Combined with compression-type reduction, results in less maintenance when crushing abrasive materials.

Today there are thousands of Williams Roll Crushers at work throughout the industry. Through the many roll designs and combinations available, a wide range of materials can be crushed to specific requirements. Just ask us for a list of some of our major applications.



- Reduction Ratio - up to 8:1
- Feed Size Range - up to 12 inches for high strength materials, larger for friable feeds
- Nominal Product Size - as small as minus 1 inch

CAPACITIES - USING MEDIUM HARD BITUMINOUS COAL

Crusher Size (in.)	Product Size Nominal Capacity Tons per Hour			Max. Feed Size	Motor HP	RPM Roll
	1"	1-1/2"	2"			
15 x 10	10 - 15	15 - 20	20 - 25	8"	5 - 7 1/2	150 - 350
15 x 20	20 - 30	30 - 40	40 - 50	8"	7 1/2 - 15	150 - 350
24 x 24	30 - 60	50 - 80	70 - 100	16"	15 - 25	60 - 220
24 x 36	50 - 90	80 - 120	100 - 140	16"	25 - 40	60 - 220
24 x 48	80 - 120	100 - 150	120 - 200	16"	40 - 60	60 - 220
30 x 20	75	100	125	18"	25 - 50	150 - 200
30 x 40	100	125	150	18"	50 - 60	150 - 200
30 x 50	150	175	200	18"	60 - 75	150 - 200
30 x 60	175	250	300	18"	75 - 100	150 - 200
36 x 40	100	150	200	20"	50 - 60	125 - 150
36 x 40	175	225	275	20"	60 - 75	125 - 150
36 x 50	200	285	350	20"	100 - 125	125 - 150

Williams Double Roll Crushers

Williams Now Manufactures the Most Advanced Double Roll Crusher Ever Built!

Roll Crushers are simple in design and construction, long lasting, economical, and versatile.





Two roll crushing surfaces operate at a fixed distance apart, so product size is much more consistent. Both oversized pieces and fines are minimized.

Double roll crushers reduce primarily through compression, although some shear is obtained with toothed rolls. Double rolls come in combinations of smooth, corrugated, and toothed designs. Double roll crushers operate most efficiently with dry, friable materials. Double roll crushers are usually used to produce a finer product at lower reduction ratios.

Williams Two Roll Crushers are built rugged for high tonnage with minimum maintenance.

Simple drive arrangement utilizes 16 ply truck tires which eliminates gears, sprockets, and chains.



Typical Feed Materials for Williams Roll Crushers		
Alundum	Coal	Rock Salt
Bauxite	Glass	Sandstone
Cement Clinker	Gypsum	Shale
Chalk	Limestone	Sulfur Ore
Cinders	Burnt Lime	Sea Shells
Clay	Petroleum Coke	Sewer Sludge Clinker

FEATURES:

Less maintenance when crushing abrasive materials because of low peripheral speed.
Reduction ratio- up to 8:1; Feed size range- up to 3" for high strength materials, larger for friable feeds; Nominal product size- as small as minus 10 mesh

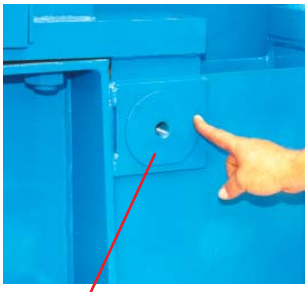


Simple Roll Adjustment
Quickly set and maintained via manual/Servo Motor Driven Screw-Jack (Patent Pending) arrangement.

Choice of Drives
Multiple V-belt drives recommended for maximum shock absorption. Shaft-mounted speed reducers optional.

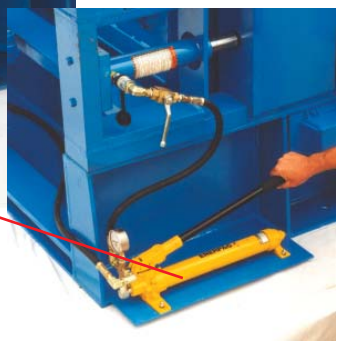


Welded Steel Frame
Heavy steel plate and welded structural shapes for rigidity and resistance to shock.



Pivot Point of the adjustable roll



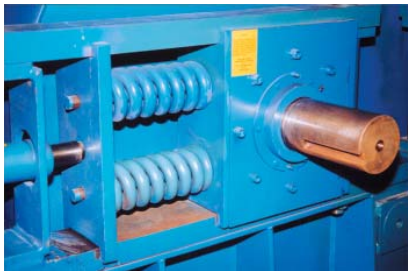


Roll Pressure Adjustment
Roll pressure adjustable through simple hydraulic system.

Heavy Duty Drive Shafts and Bearings
Oversized, heat-treated, alloy steel shafts plus self-aligning, roller-type bearings assure long life and maximum use of power.



Replaceable Roll Faces
Smooth tooth or corrugated patterns to suit feed material. Abrasion-resistant hardened alloy. Ringfetter mounted for easy replacement.



Protective Spring Releases
Heavy-duty compression springs permit movement of floating roll to pass tramp metal and other uncrushable items, thus avoiding overload & damage.

CAPACITIES - USING MEDIUM LIMESTONE

Crusher Size (in.)	Approx. Capacity in Tons per Hour at Indicated Spacing Between Rolls						Motor HP	RPM Rolls
	1/8"	1/4"	3/8"	1/2"	3/4"	1"		
11 x 10	2	4	5	8	10	16	5 - 10	100-200
11 x 20	4	8	10	16	20	32	10 - 15	100-200
20 x 18	9	18	27	36	56	72	25 - 40	50 - 100
30 x 24	13	26	32	52	64	105	40 - 60	30 - 60
30 x 40	16	28	40	55	75	110	40 - 60	30 - 60
30 x 50	18	30	45	60	90	120	40 - 60	30 - 60
36 x 40	18	30	45	60	90	120	40 - 60	30 - 60
36 x 50	20	40	60	80	125	160	60 - 75	30 - 60
36 x 60	25	50	75	100	150	200	75 - 100	30 - 60

Note: Maximum feed size is approx. 3 times the roll spacing.

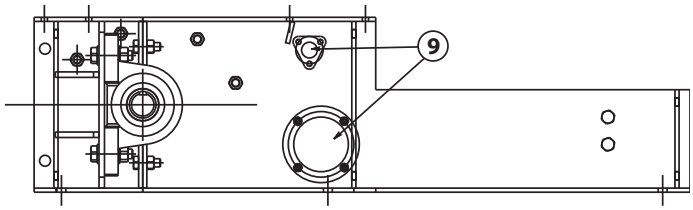
Williams Ash Crushers for Dust-Tight Operations

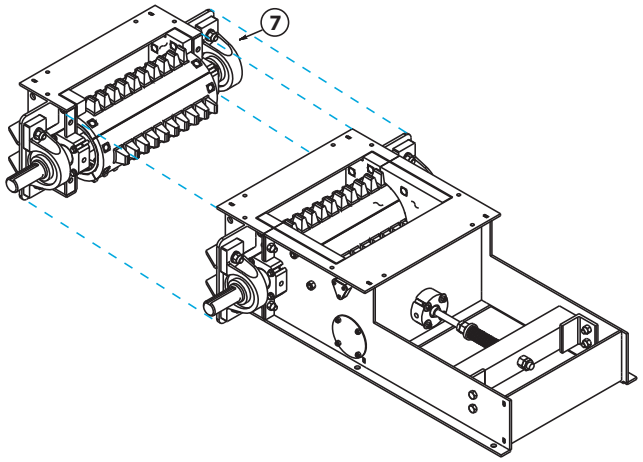


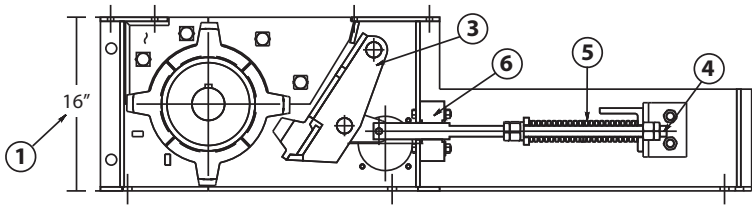


Williams Single Roll Crushers have always been built rugged for heavy duty applications. Now, we have designed a new unit that is dust-tight. This machine is perfect for applications such as crushing ash, limestone, coal, or glass where it would be expensive to have dust collection air.

The Williams 15" diameter Single Roll Crusher is known for its rugged construction, low profile, high reduction ratio, and economical cost. It is an excellent unit for dust tight operation and easy access to the rotor for maintenance.

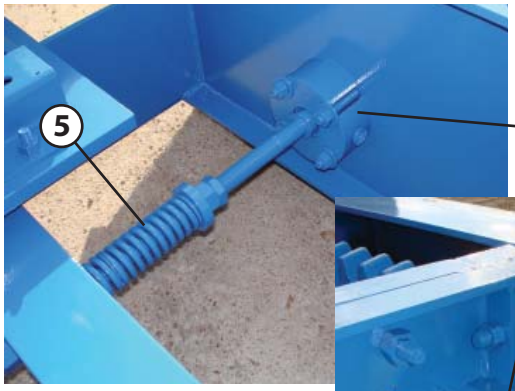




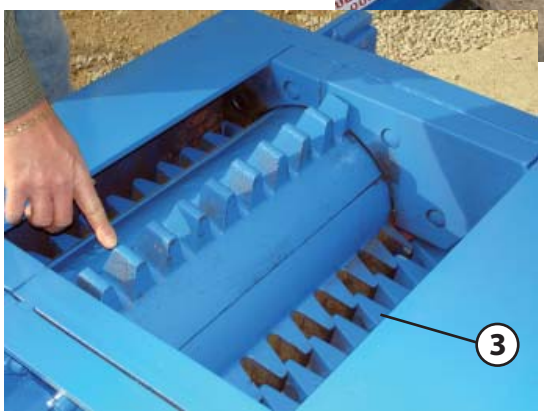


FEATURES:

1. 16" Low Profile Design
2. Dust-Tight Construction
3. Adjustable Compression Spring-Loaded Breaker Plate
4. Breaker Plate to Rotor Clearance Setting by means of External Adjustment Rod
5. Breaker Plate Pressure set by adjusting Heavy Duty Pressure Spring
6. Shaft Seals and Adjustment Rod Seal for Dust-Tight Housing
7. Horizontal removal of Rotor, Shaft Seals, and Bearing Supports allows for maintenance with out disturbing the Feed and Discharge Chutes
8. V-Belt Drive eliminates the need for a costly Reducer
9. Sealed Cover Plates for Breaker Plate Access
10. Cast Iron Internal Components rated to crush material up to 400°F

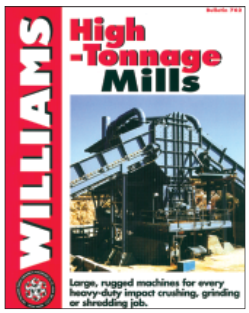






CAPACITIES - BASED ON FRIABLE MATERIALS SUCH AS COAL, BOTTOM ASH, OR GLASS						
Crusher Size (in.)	Product Size Nominal Capacity Tons per Hour			Max. Feed Size	Motor HP	RPM Roll
	1"	1-1/2"	2"			
15 x 10	10 - 15	15 - 20	20 - 25	8"	5 - 7 1/2	150 - 350
15 x 20	20 - 30	30 - 40	40 - 50	8"	7 1/2 - 15	150 - 350

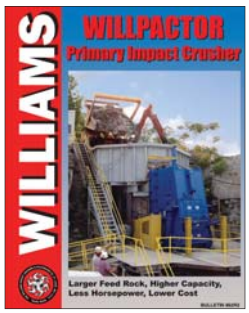
Call us to see our complete line of Williams Crushers!



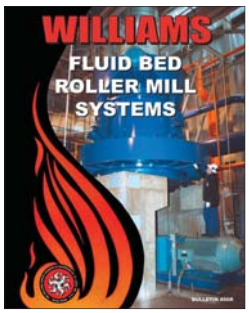
BULLETIN 762



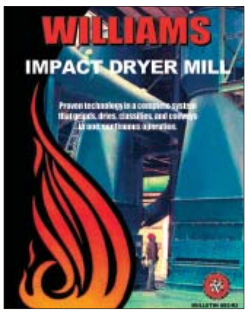
BULLETIN 766



BULLETIN 862



BULLETIN 856



BULLETIN 882

The Williams Way

At Williams, the company's mission is to provide highly personalized engineering that starts with a complete understanding of the size reduction need and ends with a machine designed and built to meet it.

Along the way, a CAD/CAM interactive graphics computer system is employed which permits three-dimensional, "on-screen" design of a part and translates it into numerical control tapes for operation of burning tables and machine tools.

Additionally, our complete test laboratory is geared to analyzing problems and evaluating answers.

Try the Williams Way. You'll like it. See your Williams Representative or contact us directly.

The Williams Roll Crusher for Your Needs

Williams maintains a complete test facility and latest technology-backed engineering department. We will make pilot runs on samples of your product and assist in the evaluation of test results. Test data are treated as a customer's confidential information.

Tests may be arranged by contacting Williams directly, or contacting your local Williams Representative.

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